

LEGO® Education SPIKE™ Prime Technical Specifications

Technic™ Large Hub



Hardware name	Technic™ Large Hub
Description	The Hub is a programmable control unit that LEGO® sensors and motors can be attached to. The Hub has an intuitive light and button interface and can work autonomously or in streaming mode.
Key features	• 5x5 LED matrix white display • Six input/output ports • Six-axis Gyro Sensor (three-axis accelerometer and three-axis gyroscope) • Three-button navigation, including light interface • Speaker • Connectivity to devices using USB and Bluetooth • MicroPython operating system • The Hub has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO Power Functions 2.0 (LPF2) for connection to LEGO sensors and motors
Hub capabilities	5x5 LED matrix white display: • 25 white LEDs • Each LED is programmable • Each LED can be dimmed in 10-step increments Input/output ports: • 6x LPF2 input/output ports (ports A, B, C, D, E, and F) • 115 kB port speed (ports E and F are prepared for “high-speed”) • Autodetection of sensors and motors • Ports can be chosen freely when pairing motors Six-axis Gyro Sensor: • Three-axis accelerometer • Three-axis gyroscope • Able to report: • Gyroscope mode (three-axis) • Accelerometer/tilt mode (three-axis) • Gestures as <i>tap</i>, <i>free fall</i>, and <i>shake</i> Three-button interface: • Center Button: • Turns the Hub on/off • Executes selected programs • RGB light LEDs inside the button that communicate system status and are programmable • Left/Right Button: • Navigates programs on the Hub Speaker: • Used for interface sounds • Maximum sound quality is 12 bit 16 KHz (mono)

The LEGO® Education SPIKE™ app may not support all hardware features and functionalities.



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Wireless connectivity:

- Enables users to connect wirelessly to smart devices (phones, tablets, laptops, etc.) and LEGO® LPF2 remotes and wireless components using Bluetooth
 - Bluetooth Classic 4.2 (BTC)
 - Bluetooth Low Energy 4.2 (BLE)
- Connectivity to smart devices (laptops, tablets, etc.) is only supported with BTC
- Programs, sounds, and firmware can be updated wirelessly (BTC only)
- Minimum range: 10 meters (line-of-sight, without obstructions, indoors)
- The Hub can have a maximum of four BLE and one Bluetooth connections to compatible wireless components
- Connection interface button on the Hub enables users to turn wireless functionality on/off
 - RGB light LED indicates connectivity status ("Bluetooth on/off," "connected," "out of range")

Power supply:

- Specially designed rechargeable battery is required
 - Charges inside the Hub via micro USB cable
 - Can be removed from the Hub without the use of tools
- RGB color LED indicates power/charging status

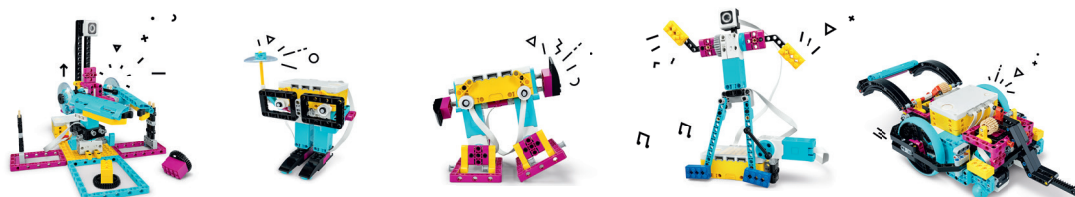
Size and weight:

- Hub Weight 63 g (without battery)
- Hub Dimensions: L88.0 x W56.0 x H32.0 mm.

System:

- The Hub is powered by a 100MHz M4 320 KB RAM 1M FLASH processor
- 32 MB of memory for programs, sound, and content
- Embedded MicroPython operating system
 - Provides an open platform for advanced users and third-parties

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LEGO® Education SPIKE™ Essential Technical Specifications

LEGO® Technic™ Small Angular Motor



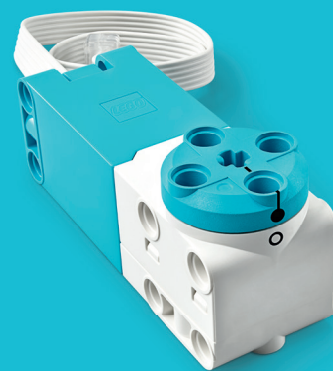
Hardware name	LEGO® Technic™ Small Angular Motor
Description	The motor is designed to function as both a motor and sensor. Using the integrated advanced Rotation Sensor, the motor can report both speed and position. The motor can also sense direct user input if the output is rotated by hand.
Key features	• Speed Sensor (measures percentage of maximum design speed) • Measures relative position (in degrees) • Measures absolute position (in degrees) • When programming in Scratch, the absolute position is 0–360 degrees • When programming in Python, the absolute position is +/- 180 degrees • Rotating disc output with crosshole and building interface • The motor has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs
Wire length	250 mm
Motor output	Voltage Range • Min: 3.3 V • Max: 6 V No Load • Torque: 0 Ncm • Speed: 110 RPM +/- 15% • Current consumption: 55 mA +/- 15% Maximum Efficiency • Torque: 1.8 Ncm • Speed: 85 RPM +/- 15% • Current consumption: 170 mA +/- 15% Stall • Torque: 5 Ncm • Speed: 0 RPM • Current consumption: 340 mA +/- 15% All performance data is based on a 5 V power supply.
Sensor input	Rotation Sensor • Resolution: 360 counts per revolution • Resolution is the number of counts the sensor makes for every full revolution (360 degrees) on the output axle • Accuracy: • Sensor input accuracy is +/- 1 degree • Motor can be controlled with an accuracy of +/- 3 degrees • Update rate: 100 Hz • Update rate is defined as the frequency at which a new sensor reading, position, and speed are available

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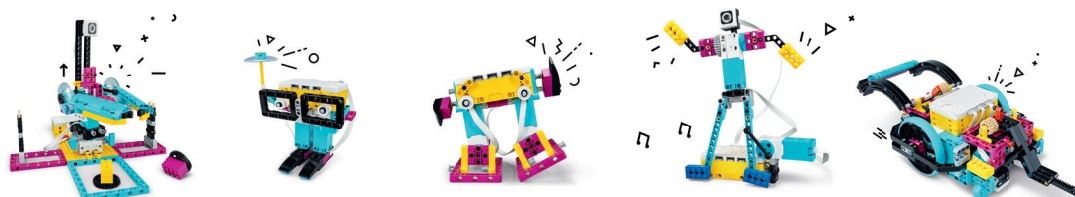
LEGO® Education SPIKE™ Prime Technical Specifications

Technic™ Medium Angular Motor



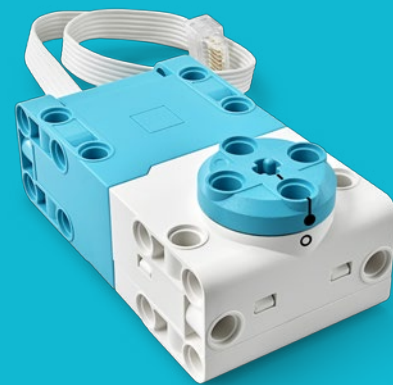
Hardware name	Technic™ Medium Angular Motor
Description	The motor is designed to function in as both a motor and sensor. With the integrated advanced Rotation Sensor, the motor can report both speed and position. The motor can also sense direct user input measures if the output is rotated by hand.
Key features	• Speed sensor (measures percentage of maximum design speed) • Relative position in degrees • Absolute position in degrees (-/+ 180 degrees) • Crosshole output on one side, rotating disc with crosshole and building interface on the other side • The motor has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO® Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs
Wire length	250 mm
Motor output	Voltage range: • Min: 5V • Max: 9V No load: • Torque: 0 Ncm • Speed: 185 RPM +/- 15% • Current consumption: 110mA +/- 15% Maximum efficiency: • Torque: 3.5 Ncm • Speed: 135RPM +/- 15% • Current consumption: 280mA +/- 15% Stall: • Torque: 18 Ncm • Speed: 0 RPM • Current consumption: 800mA +/- 15%
Sensor input	All performance data is based on a 7.2V power supply. Rotation Sensor: • <i>Resolution</i>: 360 counts per revolution • Resolution is the number of counts the sensor makes for every full revolution (360 degrees) on the output axle • <i>Accuracy</i>: $\leq \pm 3$ degrees • Accuracy is defined as the tolerances in the sensor combined with the gearbox slack • <i>Update rate</i>: 100 Hz • Update rate is defined as the frequency at which a new sensor reading, position, and speed are available

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LEGO® Education SPIKE™ Prime Technical Specifications

Technic™ Large Angular Motor



Hardware name	Technic™ Large Angular Motor
Description	The motor is designed to function in models as both a motor and sensor. With the integrated advanced Rotation Sensor, the motor can report both speed and position. The motor can also sense direct user input if the output is rotated by hand.
Key features	• Speed Sensor (measures percentage of maximum design speed) • Relative position in degrees • Absolute position in degrees (-/+ 180 degrees) • Crosshole output on one side, rotating disc with Crosshole and building interface on the other side • The motor has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO® Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs
Wire length	250 mm
Motor output	Voltage range: • Min: 5V • Max: 9V No load: • Torque: 0 Ncm • Speed: 175 RPM +/- 15% • Current consumption: 135 mA +/- 15% Maximum efficiency: • Torque: 8 Ncm • Speed: 135 RPM +/- 15% • Current consumption: 430 mA +/- 15% Stall: • Torque: 25 Ncm • Speed: 0 RPM • Current consumption: 1900 mA +/- 15%
Sensor input	All performance data is based on a 7.2V power supply. Rotation Sensor: • Resolution: 360 counts per revolution • <i>Resolution</i> is the number of counts the sensor makes for every full revolution (360 degrees) on the output axle • Accuracy: $\leq \pm 3$ degrees • <i>Accuracy</i> is defined as the tolerances in the sensor combined with the gearbox slack • Update rate: 100 Hz • Update rate is defined as the frequency at which a new sensor reading, position, and speed are available

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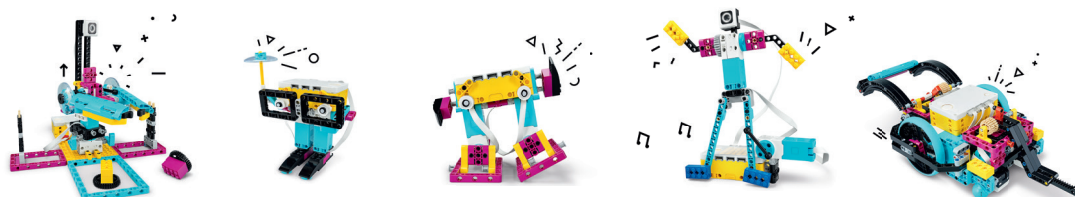
LEGO® Education SPIKE™ Prime Technical Specifications

Technic™ Color Sensor



Hardware name	Technic™ Color Sensor
Description	The sensor can detect color, reflectivity, or ambient light. The sensor can also be used as a light output.
Key features	• Color sensing (RGB, HSV, and LEGO® colors) • Reflectivity sensing (for line following) • Ambient light sensing • Emission of white light • The sensor has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs
Wire length	250 mm
Sensor sample rate	100 Hz
Sensor input	Color sensing: • Optimal reading distance: 16 mm (depending on object size, color, and surface) • Output range: • No object • White (LEGO:01; R:244, G:244, B:244) • Blue (LEGO:23; R:30, G:90, B:168) • Black (LEGO:26; R:0, G:0, B:0) • Green (LEGO:28; R:0, G:133, B:43) • Yellow (LEGO:24; R:250, G:200, B:10) • Red (LEGO:21; R:180, G:0, B:0) • Medium azur (LEGO:322; R:104, G:195, B:226) • Bright reddish violet (LEGO:124; R:144, G:31, B:118) Reflectivity sensing: • Optimal reading distance: 16 mm (depending on object size, color, and surface) • Output range: • Non-reflective/nothing = 0% • Very reflective= 100% Ambient light sensing: • Output range: • Dark = 0% • Bright = 100%
Sensor output	LED output: • Color: white - temperature 4000K • Controlled individually (3 LEDs in total) • Cannot be used while sensor is in color/light sensing mode • Output power: controllable from 0-100% increments of 1%

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LEGO® Education SPIKE™ Prime Technical Specifications

Technic™ Distance Sensor



Hardware name

Technic™ Distance Sensor

Description

The sensor can measure the distance to an object or surface using ultrasonic technology. In addition, the sensor has a light output around the “eyes,” which is divided into four segments that can be activated individually. The back of the sensor can be removed and used as “break-out” access to the LPF2 wired platform for third-party suppliers and advanced users.

Key features

Distance Sensor:

- Distance Sensing from 50 to 2000 mm
- Fast distance sensing from 50 to 300 mm
- White light output around the sensor “eyes” divided into four segments – two upper and two lower segments

Back of the sensor (break-out):

- Can be detached from the front of the sensor by removing two screws
- Provides access to an 8-pin female header that allows direct and easy access to the LPF2 system for adding third-party sensors and boards

The sensor has a Technic build geometry that allows for versatile building and easy integration into models

Connector type

LEGO® Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs

Wire length

250 mm (fixed to sensor)

Sensor sample rate

100 Hz (with ultrasonic function)

Sensor input

Distance sensing:

- Range total: 50-2000 mm +/- 20 mm
- Entrance angle: +/- 35 degrees (varies according to the distance)
- Sensor output resolution: 1 mm

Fast Distance sensing:

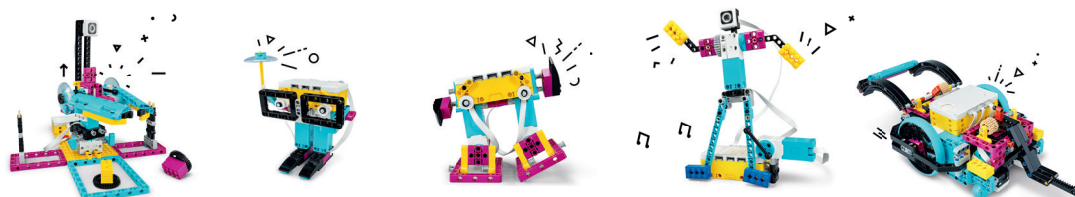
- Range total: 50-300 mm +/- 15 mm
- Entrance angle: +/- 35 degrees (varies according to the distance)
- Sensor output resolution: 1 mm

Sensor output

LED output:

- Color: white 4000K
- Controlled individually (4 segment LEDs in total)
- Output power: controllable from 0-100% in increments of 1%

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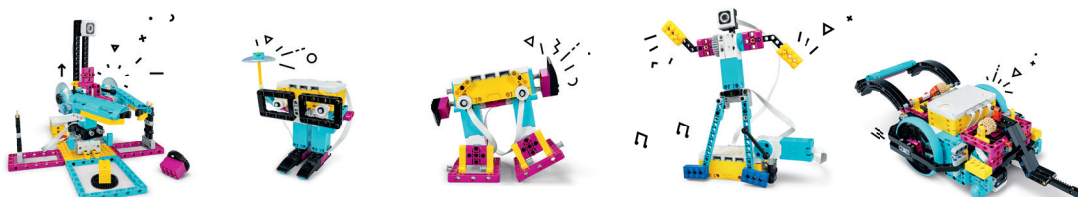
LEGO® Education SPIKE™ Prime Technical Specifications

Technic™ Force Sensor



Hardware name	Technic™ Force Sensor
Description	The sensor can detect simple touch and measure force.
Key features	• Touch sensing • Tap sensing • Force sensing • The sensor has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO® Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs
Wire length	250 mm (fixed to sensor)
Sensor sample rate	100 Hz
Sensor input	Touch sensing: • Activation zone: 0-2 mm • Threshold point is defined in firmware 1 mm +/- 0.5 mm • Activation force: 0.5-1.0 newton +/- 10% • Sensor output is binary (1=activated or 0=not activated) Tap sensing: • Activation zone: 0-2 mm • Threshold point is defined in firmware 1 mm +/- 0.5 mm • Activation force: 0.5-1.0 newton +/- 10% • Sensor data output: 0-3 • Applicable to the following: • Single tap • Quick tap • Press and hold Force sensing: • Activation zone: 2-8 mm • Activation force: 2.5-10 newtons • Sensor output resolution: 0.1 newton steps • Sensor output is limited to a fixed max output value: 10 newtons • Sensor output accuracy: +/- 0.65 newton Force-filter sensing (high speed “peak” sensing): • Like the force sensing mode • Internal sensor sample rate: 1kHz • Keeps the maximum value until reset by the Hub firmware or powered down

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LEGO® Education SPIKE™ Essential Technical Specifications

LEGO® Technic™ 3×3 Color Light Matrix



Hardware name	LEGO® Technic™ 3×3 RGB Light Matrix
Description	The RGB Light Matrix is a programmable light matrix with the capability to turn on each segment separately and in different colors.
Key features	• 3X3 RGB LED matrix display • 9 RGB LEDs • Each LED is programmable in 10 different colors • Each LED can be dimmed in 10-step increments • The transitions between 2 frames can be instant (standard), faded, or swiped • The Light Matrix has a Technic build geometry that allows for versatile building and easy integration into models
Connector type	LEGO® Power Functions 2.0 (LPF2) for connection to LEGO Smarthubs.
Wire length	250 mm

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